Smart Parking: Transforming into a data-intelligence business

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About Smart Parking

Operating from New Zealand, Australia, and the United Kingdom, Smart Parking provides end-to-end smart parking management/smart city solutions for cities and businesses globally.

Industries: Technology

Location: New Zealand, Australia, United Kingdom

App Engine

Google Cloud results

- Reduced smart parking/smart

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city IoT installation and operational support effort by more than half

- Enabled development of a Smart Cloud IoT platform in just four months
- Democratised data access and use across the organisation

Cloud IoT Core (https://cloud.google.com/iot-core/)

Cloud Functions (https://cloud.google.com/functions/)

Cloud Pub/Sub (https://cloud.google.com/pubsub/)

Dataflow (https://cloud.google.com/dataflow/)

App Engine (https://cloud.google.com/appengine/)

BigQuery (https://cloud.google.com/bigquery/)

Looker Studio (https://marketingplatform.google.com/about/data-studio/)

Vision AI (https://cloud.google.com/vision/)

AI Platform (https://cloud.google.com/ai-platform/)

Smart Parking (https://www.smartparking.com/)'s core product is a sensor-based system called SmartPark, used in environments such as shopping centres, airports, commercial parking operations, universities, and municipal streets.

Smart Parking has deployed over 50,000 sensors worldwide to support parking systems. It estimates at least 70% of production-scale and currently operating smart parking environments globally are using its technologies.

"Our products include ground sensors that use an advanced combination of infrared and magnetic technologies to reliably register a vehicle's arrival and communicate to a gateway," says Brian Granatir, Technical Team Lead, Smart Parking. This enables operators to gain live insights into parking environment usage and manage capacity. They can also provide automated guidance and signage to tell customers how many vacant spaces are available on each level of a parking structure or street area of a city.

Additional functionality enables parking operators to identify whether vehicles parked in limited-time spots have overstayed. The operators may notify inspectors who can undertake enforcement actions such as issuing infringement notices.

Another Smart Parking mobile product enables parking operators to photograph licence plates to identify vehicles that have overstayed. The business is also developing applications that enable users to view the number of parking spots available in nearby parking structures.

Smart Parking is extending its operations into smart city developments. The business views this as a natural progression as its parking systems establish a powerful common foundation from which to deliver many other services. "Parking is just part of a very complex ecosystem that requires sophisticated interoperability and generalised analytics," John Heard, Chief Technology Officer, Smart Parking, explains.

"The biggest trend in the space is towards the Internet of Things (IoT)–the integration of a large number of generic and specialised devices," Heard adds. "This has led us to shift from the idea of our devices being the centre of the universe. Instead, we put data intelligence at the core."

> "By running on Google, we were able to develop our SmartCloud Platform at an incredible pace. We built the core infrastructure in under four months."
>
> *—Brian Granatir, Technical Team Lead, Smart Parking*

In October 2016, Smart Parking decided to develop a SmartCloud platform that would enable cities to manage and react to information from a connected network of IoT devices. The business wanted to start by rolling out the platform based on an event-driven architecture that could connect any device that sends or receives a sequence of data events to existing Smart Parking customers.

The platform would enable Smart Parking to address a wide range of highly complex and interrelated interactions. “Internally, we use the example that if everyone leaves a parking structure at 3pm, what will be the impact on traffic?” Heard says. “So, how can we correlate multiple factors to answer a larger question?”

Heard’s team elected to deploy SmartCloud on Google Cloud (https://cloud.google.com/), due primarily to its support for sophisticated big data management and machine learning in a serverless architecture. Furthermore, Google technologies were proven to operate at scale and Google Cloud had baked-in best practices that could manage internet-scale datasets in near real time.

“By running on Google, we were able to develop SmartCloud at an incredible pace,” Granatir says. “We built the core infrastructure in under four months.” Furthermore, Google Cloud enables the SmartCloud platform to operate at metropolis scale, combining streams of distributed data to deliver an unprecedented view of the events and interactions taking place in a city.

Smart Parking now uses a wide range of Google Cloud services to deliver SmartCloud. Google Cloud Dataflow (https://cloud.google.com/dataflow/) provides processing and Google BigQuery (https://cloud.google.com/bigquery/) provides management and analytics of big data. Google

Cloud Pub/Sub
 (https://cloud.google.com/pubsub/docs/overview)
provides real-time messaging and Google Cloud Functions (https://cloud.google.com/functions/) enables a serverless environment to enable delivery of streaming data. Google Cloud IoT Core (https://cloud.google.com/iot-core) connects, manages, and ingests data from Smart Parking devices while Google Cloud Identity & Access Management (https://cloud.google.com/iam/) enables Smart Parking to control access to information. Google Cloud Pub/Sub and Google Cloud Functions also support the event-driven architecture.

Installation time and operational burdens cut by at least half

Using Google Cloud IoT Core has cut Smart Parking's parking and smart city system installation times by at least half and eliminated the need to build a registration system, a device authorisation system, and a feedback loop for each project. This delivers a streamlined and simpler methodology for site installations, which flows on to a dramatic decrease in the operational burdens involved in monitoring and maintaining deployments of all scales.

Using Google Cloud has revolutionised Smart Parking's software development processes. "We're building and deploying systems at a speed I could never have imagined before Google Cloud," Granatir says. Furthermore, using Google Cloud has made data accessible to anyone within the Smart Parking business, and this is being extended to customers via SmartCloud information services.

> “Because of Google Cloud and its pricing, we don’t walk in the shadow of giants, we ride on their shoulders.”
>
> *—John Heard, Chief Technology Officer, Smart Parking*

This democratised approach has already realised considerable benefits. “One of our project managers was experimenting with the Looker Studio (https://marketingplatform.google.com/about/data-studio/) service for visualising data and was able to make a query about utilisation efficiency,” Heard adds. “We found that simply changing the maximum stay time by a small amount could hugely impact the percentage of parking utilisation and, therefore, throughput.

"This is an illustration of a significant insight by someone who simply had a hunch and quick access to mountains of live and accurate data."

Transition to a data-intelligence company

Using Google Cloud has enabled Smart Parking to chart a path from a sales and maintenance organisation to a smart cities data-intelligence company. "How? Simply by providing amazing services that focus on massive data volumes and sharing documentation that helps us leverage best practices," Heard says. "Because of Google Cloud and its pricing, we don't walk in the shadow of giants, we ride on their shoulders."

Smart Parking is now looking to further extend its use of Google services to add value to its smart parking and future smart city clients. "With Google Cloud Machine Learning Engine (https://cloud.google.com/ml-engine/), we could gain the ability to identify trends across every customer parking site and make recommendations to operators," Heard says. "For example, we could provide data-driven insight and advice to operators about how to adjust their parking spot time limits and guidance about how to transform throughput and convenience while increasing revenue."

Heard concludes, "As a result of being able to dramatically decrease our traditional development burdens by using Google Cloud, we now have a number of exciting new capabilities we have been able to bring in to our roadmap and expand the journey we are able to open up for our customers as well.”